

Urban Rooftop Organic Farming Platform

AI-Powered Real-Time Crop Management with RAG & Computer Vision



Presented By: Neil Tran



Abstract

Urban agriculture faces critical barriers: rooftop space constraints, lack of crop-specific expertise, inadequate disease diagnosis tools, and absence of organic-compliant guidance tailored to vertical environments. This project proposes an AI-powered urban rooftop organic farming platform that integrates Retrieval-Augmented Generation (RAG), computer vision, and deep learning to deliver intelligent, real-time crop guidance. The system bridges the gap between novice urban farmers and expert agronomic knowledge — enabling disease detection, lifecycle tracking, and organic-safe recommendations through a unified, accessible interface.





1. Lack of Crop-Specific Guidance

1. Disease Diagnosis Difficult

1. Lifecycle Tracking Gaps

1. Organic Recommendation

1. Rooftop Environment Constraints

Problem Statement

Urban rooftop organic farming presents unique and complex challenges that existing agricultural platforms fail to address. Farmers lack access to crop-specific, rooftop-adapted guidance tailored to organic practices.

Disease diagnosis remains reactive and inaccurate without real-time AI-powered visual analysis. Lifecycle tracking is fragmented, leaving growers without actionable stage-by-stage insights. Organic-compliant nutrient and pest management recommendations are scarce and often generic. Rooftop constraints — including load limits, drainage, microclimate variation, and sun exposure — are rarely factored into conventional farming tools.



Proposed Solution

Our platform delivers an integrated AI-powered solution for urban rooftop organic farming, combining interactive crop project management, real-time lifecycle tracking, multimodal plant disease diagnosis, and a RAG-powered organic advisory system to guide growers at every stage.



**Interactive crop
projects**



**Real-time lifecycle
timeline**



**AI Plant Disease
Doctor**



**RAG-powered organic
advisor**

System Architecture & Workflow

1



User Input Layer: The user submits crop project data or uploads a plant image via the platform interface. The crop engine parses project metadata while the computer vision model processes the image for disease or growth stage analysis.

2



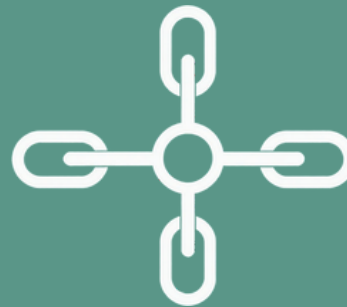
Processing & Output Layer: RAG retrieval queries the Pinecone vector store using Gemini embeddings. The AI advisor synthesizes retrieved context with LLM reasoning to generate actionable, organic-only recommendations stored in the SQLite database.

Technical Stack

The platform is built on a layered technology stack combining cutting-edge AI models, vector retrieval, orchestration frameworks, and lightweight local storage — each component selected for performance, scalability, and organic farming domain precision.



Gemini
Embeddings &
Multimodal LLM



LangChain LCEL
Orchestration



Pinecone Serverless
Vector DB



SQLite + Python Crop
Engine



AI & RAG Algorithms

AI Embedding in similarity search

1

Embedding & Similarity Search

Gemini generates 3,072-dimensional dense vector embeddings for all crop knowledge documents, enabling rich semantic representation of agronomic content.

Cosine similarity measures the angular distance between query and document vectors, ensuring semantically relevant retrieval regardless of exact keyword matches.

HNSW (Hierarchical Navigable Small World) Approximate Nearest Neighbor indexing in Pinecone Serverless enables sub-millisecond vector search at scale across the full knowledge base.

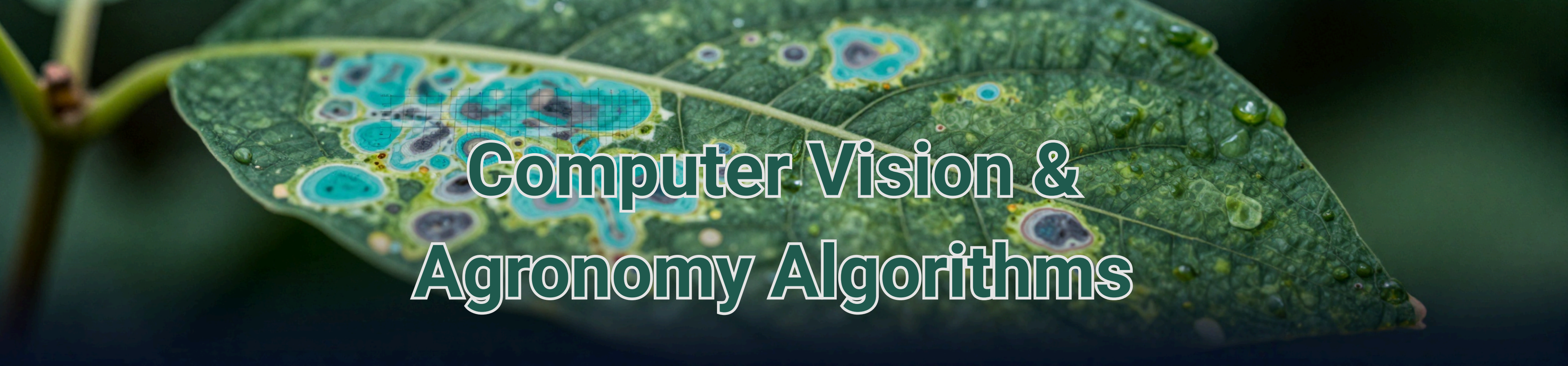
2

Chunking & Grounded Prompt Filtering

Recursive character chunking with configurable overlap splits source documents into semantically coherent segments, preserving contextual continuity across chunk boundaries for accurate retrieval.

Overlapping windows ensure no critical agronomic context is lost at chunk boundaries, improving RAG answer completeness.

Grounded prompt filtering constrains LLM responses strictly to retrieved context, preventing hallucination and ensuring all crop recommendations are evidence-based and organically validated.



Computer Vision & Agronomy Algorithms

1

Computer Vision & Disease Analysis

Multimodal vision model processes plant images for pathology feature extraction, identifying lesion patterns, discoloration, and structural anomalies.

Disease classification engine categorizes detected conditions across 50+ common crop diseases with confidence scoring. Severity scoring assigns a 0–100 scale rating to quantify infection spread and urgency, enabling prioritized treatment recommendations.

Real-time inference pipeline delivers diagnosis results within seconds of image submission.

2

Agronomy & Organic Scheduling Algorithms

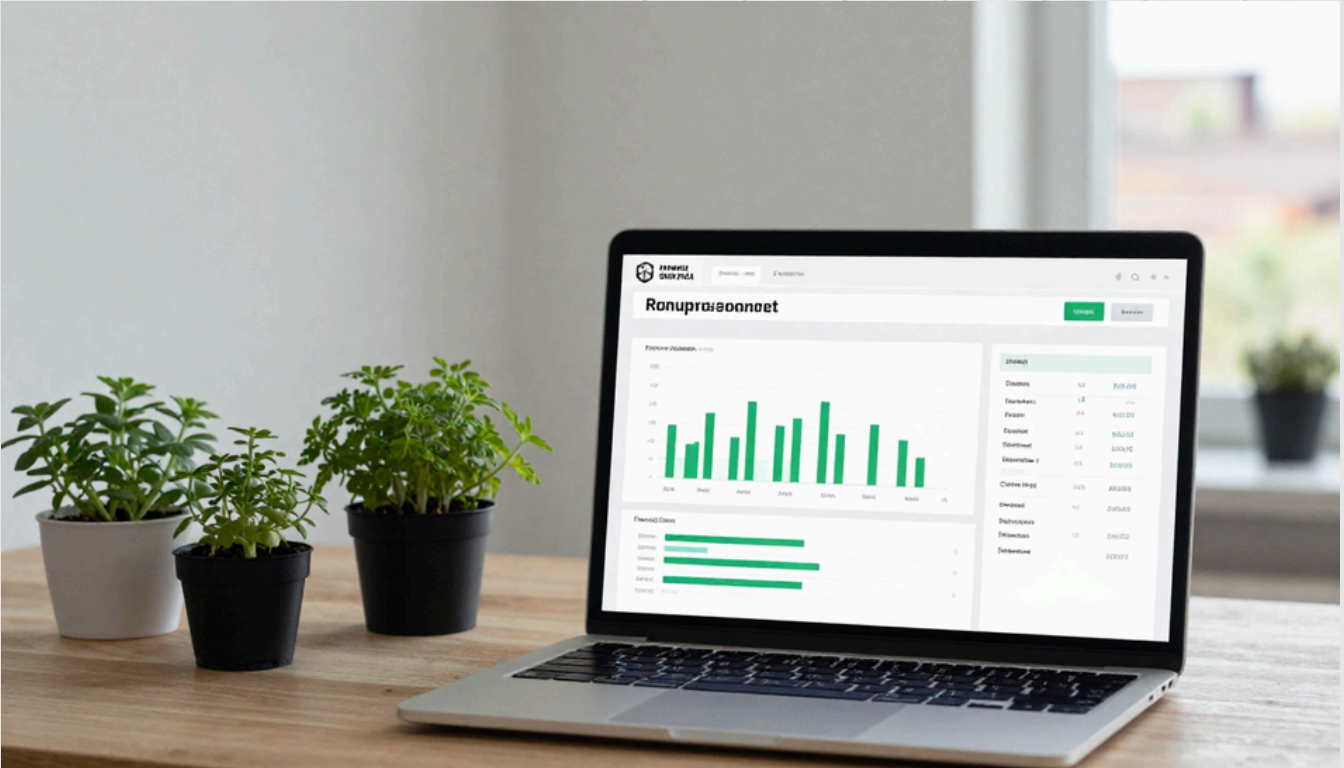
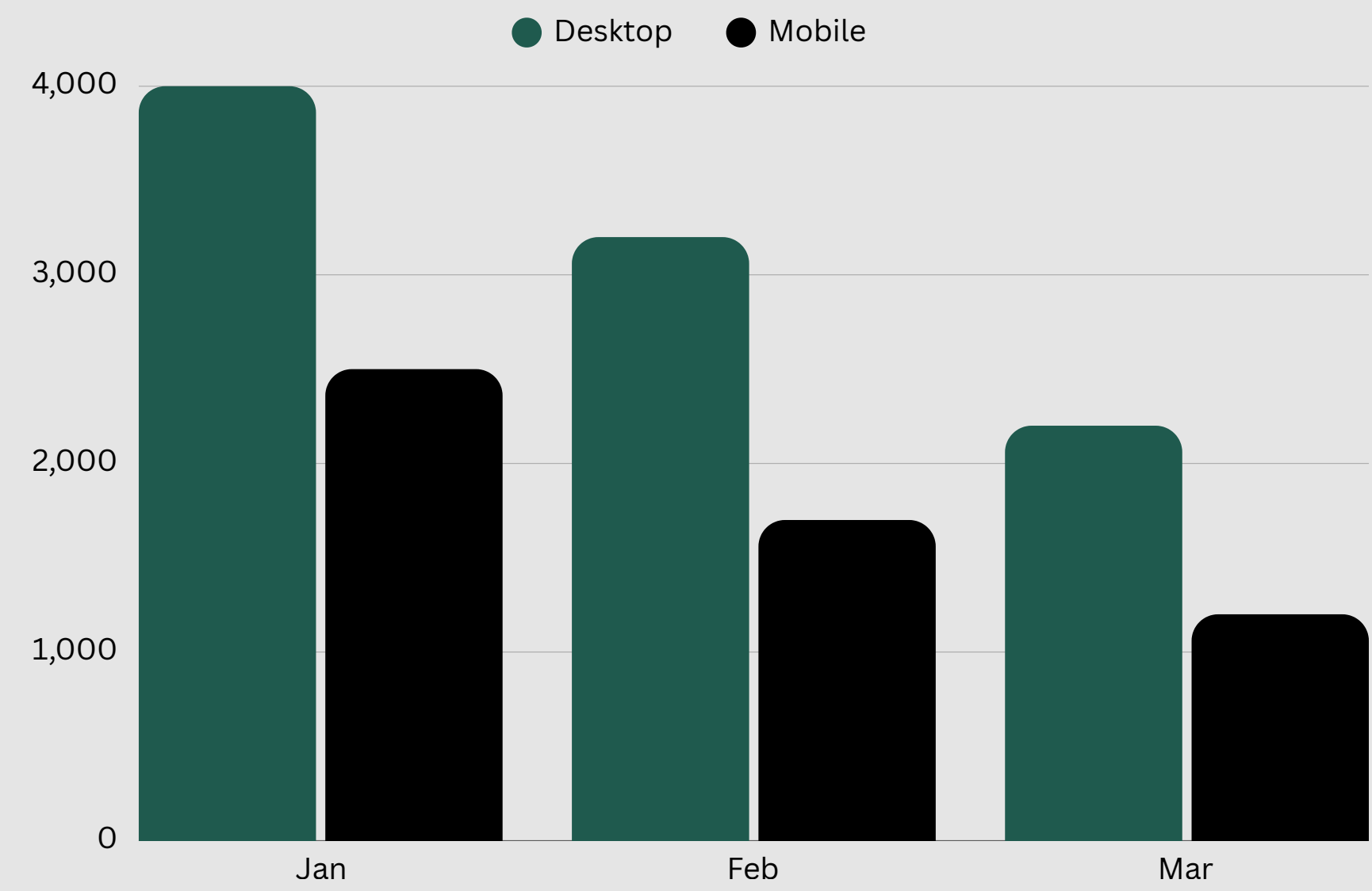
Phenological stage detection tracks crop lifecycle phases from germination to harvest, dynamically adjusting care protocols at each growth stage.

Dynamic water allocation algorithm computes irrigation schedules based on crop age, stage, rooftop microclimate, and seasonal factors to minimize waste.

Organic nutrient scheduling generates fertilization plans using only certified organic inputs, calibrated to soil health indicators and plant demand curves.

All agronomy outputs are safety-aware, accounting for rooftop load limits and urban environmental constraints.

Database & Benefits



Database Architecture: B-tree indexing enables fast crop/project lookups; SQLite WAL mode ensures concurrent read-write access for real-time tracking. Crop age formula calculates dynamic growth stages automatically.

Key Benefits: Real-time crop lifecycle guidance, organic-only compliance enforcement, AI-driven yield improvement recommendations, and seamless project status tracking deliver measurable value to urban rooftop farmers.



Conclusion & Future Scope

This platform delivers an integrated AI + agronomy solution combining Gemini LLM, RAG retrieval, computer vision disease diagnosis, and a Python crop engine — purpose-built for urban rooftop organic farming.

Key achievements include real-time lifecycle tracking, AI Plant Disease Doctor, organic-only recommendations, and safety-aware rooftop guidance. The system demonstrates the viability of AI-driven precision agriculture in constrained urban environments.

Future enhancements: IoT sensor integration for soil and microclimate data, live weather API feeds for adaptive scheduling, expanded plant disease datasets for improved diagnostic accuracy, and predictive analytics for yield forecasting and proactive intervention.